

A Tensorization Obstruction to a Scalar Closure-Defect Repair for a Union-Closed Entropy Functional

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Abstract

We analyze a scalar repair proposed for a Shapley-averaged entropy functional arising in an approach to the union-closed sets conjecture. For a finite set family $\mathcal{F}$, let $A_+(\mathcal{F})$ combine the iid OR-entropy and an optimized one-sided causal Bellman coupling, and let $\varepsilon_V(\mathcal{F})$ be the probability that the union of two independent uniform members leaves $\mathcal{F}$. The question was whether $-A_+(\mathcal{F})/\varepsilon_V(\mathcal{F})$ is universally bounded over cap-2/5, Reimer-admissible families. We prove that it is not. A 45-member normalized family on seven coordinates has a certified value $A_+ < -7/250$. Both iid and optimized causal costs are additive under Cartesian products, whereas closure success is multiplicative. Its k -fold powers are normalized and preserve the cap and Reimer conditions, but their repair ratios exceed $7k/250$. A standalone 256-bit Arb verifier certifies the sharper base value, directly checks the 2,025-member square, and evaluates the actual square Bellman recursion for two contrasting global orders. Independently, a standard-library 160-bit dyadic checker relaxes every nondegenerate action interval to $[0, 1]$ and still proves $A_+ < -1/40$, without clamp decisions or Arb. These certificates rule out every fixed scalar multiple of this ordered-pair defect. They neither produce a union-closed counterexample nor settle a separate small-defect stability question.

1 Problem and definitions

Let $\mathcal{F} \subseteq 2^{[n]}$ be a nonempty finite simple family, $m = |\mathcal{F}|$, and let all random members be uniform on $\mathcal{F}$. The admissibility conditions are

$$\max_{i \in [n]} |\{A \in \mathcal{F} : i \in A\}| \leq \left\lfloor \frac{2m}{5} \right\rfloor, \quad (1)$$

$$\sum_{A \in \mathcal{F}} |A| \geq \frac{m \log_2 m}{2}. \quad (2)$$

A family is called normalized here when every coordinate column is nonzero and the columns are pairwise distinct. Normalization is not required in the supremum below, but all members of our sequence satisfy it.

Fix a coordinate order π . At coordinate i and predecessor prefix u , write

$$p_i(u) = \mathbb{P}[X_i = 1 \mid X_{\text{pred}_\pi(i)} = u].$$

For two independent prefixes u, v , define

$$Q_\pi(\mathcal{F}) = \sum_i \mathbb{E}_{u,v} h(p_i(u) + p_i(v) - p_i(u)p_i(v)),$$

where h is binary entropy in bits, and put $Q(\mathcal{F}) = \mathbb{E}_\pi Q_\pi(\mathcal{F})$.

For the causal term, at a pair of prefixes let $p = p_i(u)$ and $r = p_i(v)$. Set

$$s^*(p, r) = \max\{p, r, \min(p + r, 1/2)\}, \quad U(p, r) = \min(1, p + r).$$

A one-sided action chooses $s \in [s^*, U]$, with next-bit probabilities

$$P_{00} = 1 - s, \quad P_{10} = s - r, \quad P_{01} = s - p, \quad P_{11} = p + r - s.$$

The fixed-order Bellman value $C_{+, \pi}(\mathcal{F})$ is the maximum expected sum of $h(s)$ over all such causal policies. We average over all orders to obtain $C_+(\mathcal{F})$. The coefficient is the fixed, dimension-independent rational

$$\alpha = 0.0356069 = \frac{356069}{10^7},$$

and the functional is

$$A_+(\mathcal{F}) = (1 - \alpha)Q(\mathcal{F}) + \alpha C_+(\mathcal{F}) - \log_2 m. \quad (3)$$

The ordered-pair closure defect is

$$\varepsilon_V(\mathcal{F}) = \mathbb{P}_{X, Y \sim \text{Unif}(\mathcal{F})}[X \cup Y \notin \mathcal{F}]. \quad (4)$$

The Gate B quantity is

$$c_{\text{cl}}^* = \sup_{\substack{\mathcal{F} \text{ satisfies (1), (2)} \\ A_+(\mathcal{F}) < 0}} \frac{-A_+(\mathcal{F})}{\varepsilon_V(\mathcal{F})}. \quad (5)$$

The quotient is interpreted in the extended nonnegative reals: an admissible family with $A_+(\mathcal{F}) < 0$ and $\varepsilon_V(\mathcal{F}) = 0$ has value $+\infty$; otherwise the quotient is taken only at positive defect. Our construction has strictly positive defect, so the theorem is independent of this convention. No upper bound on ε_V occurs in (5).

2 Certified negative base

Partition the seven coordinates as $\{5, 6\} \sqcup \{0, 1, 2, 3, 4\}$ and let $\mathcal{B}$ be the union of the seven weight cells

$$(0, 1), (0, 3), (1, 0), (1, 2), (1, 5), (2, 0), (2, 1). \quad (6)$$

The exact row-mask list is stored in the accompanying machine-readable candidate. Direct counting gives

$$|\mathcal{B}| = 45, \quad d_i = 18 \quad (0 \leq i < 7), \quad \sum_{A \in \mathcal{B}} |A| = 126.$$

Thus the cap holds with equality. The exact Reimer threshold is 124. The columns are active and pairwise distinct. There are exactly 1,600 missing ordered join pairs, and hence

$$\varepsilon_V(\mathcal{B}) = \frac{64}{81}, \quad 1 - \varepsilon_V(\mathcal{B}) = \frac{17}{81}. \quad (7)$$

The family is not union-closed: rows 1 and 2 occur but row 3 does not.

Proposition 1 (Certified base). *For the functional (3),*

$$A_+(\mathcal{B}) \leq -0.0286491867944683178160269789255044875 \dots < -\frac{7}{250}. \quad (8)$$

The proposition is a finite computational lemma. A standalone checker reconstructs $\mathcal{B}$ from the cell predicate in (6), verifies that it equals the stored row list, rebuilds the exact constraints and the 240-element automorphism group, reduces the 5,040 coordinate orders to 21 equal-size orbits, and produces a rigorous upper certificate for the one-sided Bellman maximum with 256-bit Arb arithmetic. At each of 982 clamp-ambiguous states the unconstrained concave maximum is retained, which upper-bounds the constrained maximum. The checker imports no implementation module from the original discovery code.

Proposition 2 (Independent clamp-free certificate). *There is a standard-library-only, outward-rounded dyadic certificate that proves*

$$A_+(\mathcal{B}) < -0.0271742911034863966862278304907 < -\frac{1}{40}.$$

Proof. For a fixed order, define a relaxed backward value W as follows. At a nondegenerate state, replace the feasible one-sided action interval by all of $[0, 1]$. Given child values W_{ac} , put

$$D = -W_{00} + W_{10} + W_{01} - W_{11}.$$

The unrestricted concave maximum is

$$W_{00} - rW_{10} - pW_{01} + (p + r)W_{11} + \log_2(1 + 2^D).$$

At a degenerate state the action is forced and the boundary recurrence is used. Backward induction gives $C_{+,\pi} \leq W_\pi$: before enlarging the action interval, all four feasible transition coefficients are nonnegative, so replacing each true child by an upper child preserves the inequality; maximizing over a superset can only increase it.

The checker reconstructs $\mathcal{B}$ from (6), computes Q directly by the predecessor-subset Shapley formula, and exactly partitions all 5,040 orders into the 21 equal orbits of the verified 240-element automorphism group. It obtains the outward upper endpoints

$$Q(\mathcal{B}) < 5.447354977380321988160699884419, \quad \mathbb{E}_\pi W_\pi < 5.933885055704082998831935023544,$$

which imply (2). Natural logarithms use a rational atanh series with an exact tail bound; exponentials use range reduction, a rational Taylor tail bound, and repeated squaring. Thus no floating-point library or Arb operation enters this second certificate. $\square$

3 Tensorization

For families on disjoint coordinate blocks define

$$\mathcal{F} \boxtimes \mathcal{G} = \{A \cup (n + B) : A \in \mathcal{F}, B \in \mathcal{G}\}.$$

Lemma 3 (Closure success).

$$1 - \varepsilon_V(\mathcal{F} \boxtimes \mathcal{G}) = (1 - \varepsilon_V(\mathcal{F}))(1 - \varepsilon_V(\mathcal{G})). \quad (9)$$

Proof. A product join lies in the product family if and only if both block joins lie in their respective families. Under two independent uniform product rows, the two events are independent. $\square$

Lemma 4 (Cost tensorization).

$$Q(\mathcal{F} \boxtimes \mathcal{G}) = Q(\mathcal{F}) + Q(\mathcal{G}), \quad C_+(\mathcal{F} \boxtimes \mathcal{G}) = C_+(\mathcal{F}) + C_+(\mathcal{G}). \quad (10)$$

Proof. Fix a global coordinate order π and write π_F, π_G for its induced relative block orders. A uniform product row has independent blocks. Conditioning on predecessor coordinates in the other block therefore does not alter a within-block conditional-one probability. Each iid local term is the corresponding factor term, giving

$$Q_\pi(\mathcal{F} \boxtimes \mathcal{G}) = Q_{\pi_F}(\mathcal{F}) + Q_{\pi_G}(\mathcal{G}).$$

For the optimized causal term, split every pair of product prefixes into its two block states. We prove by backward induction on the remaining global suffix that the Bellman value decomposes as $V = V_F + V_G$. The terminal case is immediate. Suppose the next coordinate belongs to the F block. Its conditional marginals, action interval, and transition probabilities depend only on the F prefixes. By induction, every child value is its F -child value plus the same unchanged constant V_G . For every feasible action,

$$\sum_{a,c} P_{ac} = (1-s) + (s-r) + (s-p) + (p+r-s) = 1,$$

and hence

$$\max_s \left\{ h(s) + \sum_{a,c} P_{ac}(V_{F,ac} + V_G) \right\} = V_G + \max_s \left\{ h(s) + \sum_{a,c} P_{ac}V_{F,ac} \right\}.$$

The common shift also cancels from $-V_{00} + V_{10} + V_{01} - V_{11}$, so it cannot change the optimizer. The remaining maximization is exactly the F Bellman step. The other block is symmetric. In particular, a policy cannot gain by using other-block prefixes: their continuation contribution cancels from the Bellman slope and hence from the optimizer. Thus

$$C_{+,\pi}(\mathcal{F} \boxtimes \mathcal{G}) = C_{+,\pi_F}(\mathcal{F}) + C_{+,\pi_G}(\mathcal{G}).$$

A uniform global order induces uniform relative orders in each block. Averaging proves (10). $\square$

Corollary 5.

$$A_+(\mathcal{F} \boxtimes \mathcal{G}) = A_+(\mathcal{F}) + A_+(\mathcal{G}). \quad (11)$$

Proof. Combine (10) with $\log_2 |\mathcal{F} \boxtimes \mathcal{G}| = \log_2 |\mathcal{F}| + \log_2 |\mathcal{G}|$ in (3). $\square$

4 Unbounded repair ratio

Let $\mathcal{F}_k = \mathcal{B}^{\boxtimes k}$. Then $|\mathcal{F}_k| = 45^k$ and the dimension is $7k$. Every coordinate occurs in

$$18 \cdot 45^{k-1} = \frac{2}{5} 45^k$$

rows, so the cap holds with equality. The total incidence is $126k45^{k-1}$ and the average row size is $14k/5$. Since

$$45^5 = 184,528,125 < 268,435,456 = 2^{28},$$

we have $\log_2 45 < 28/5$, and (2) holds strictly for every k . Activity and separation within a block are inherited from $\mathcal{B}$; columns in different blocks are distinct because base columns take both values and blocks vary independently. Hence every $\mathcal{F}_k$ is normalized.

Equations (7), (9), and (11) give

$$\varepsilon_V(\mathcal{F}_k) = 1 - \left(\frac{17}{81} \right)^k, \quad A_+(\mathcal{F}_k) = kA_+(\mathcal{B}).$$

Theorem 6 (Gate B). *The quantity in (5) is infinite.*

Proof. By (8),

$$\frac{-A_+(\mathcal{F}_k)}{\varepsilon_V(\mathcal{F}_k)} > \frac{7k/250}{1 - (17/81)^k} > \frac{7k}{250} \rightarrow +\infty.$$

Every member of the sequence is admissible, so the supremum is infinite. $\square$

5 Computational controls

The proof uses computation only to establish the strictly negative finite base. The standalone verifier additionally performs the following adversarial checks.

1. It explicitly constructs the 2,025-row square and counts 3,920,000 missing ordered joins, obtaining $\varepsilon_{\vee}(\mathcal{F}_2) = 6272/6561 = 1 - (17/81)^2$.
2. It evaluates the actual square-family Bellman recursion in an exact factored-fiber representation for consecutive and alternating global orders. Both Arb enclosures agree with twice the base-order optimum, with residual radius below 1.4×10^{-70} . The alternating computation visits 2,985,831 states and encounters 405,641 ambiguous clamps, retaining the unconstrained upper bound at each.
3. It records $\alpha = 356069/10^7$ independently of dimension and checks the cap, incidence, Reimer witness, and exact defect arithmetic for powers $1 \leq k \leq 8$. The all- k conclusion is the proof above, not extrapolation from these instances.
4. A separate regression suite attacks the fixed-order product identity for all 225 ordered pairs of nonempty two-coordinate families and all 24 global orders of each product: 5,400 interleaved Bellman comparisons. It checks the iid and one-sided identities with the legacy float implementation and the defect identity exactly. It also compares the independent Arb and legacy float Bellman evaluators and brute-checks the $n = 8$ order quotient.
5. A non-factored audit materializes the same square as a plain list of 2,025 fourteen-bit masks and runs generic prefix-fiber evaluators on it, with no product-aware shortcut, for five declared global orders: consecutive, block-swapped, reversed, alternating, and a seeded interleaving. The directly computed $C_{+,\pi}$ and Q_{π} match the induced block sums to float64 rounding, worst gaps 8.9×10^{-16} and 3.6×10^{-15} , while the values take three distinct magnitudes.
6. A resumable falsification search attacks additivity on 292 ordered products, 205 with unequal block dimensions, including dead columns, duplicate columns, singletons, full power sets, middle layers, and seeded random pairs, over every global order of each product. It found no counterexample; the worst gap is 1.8×10^{-15} and the exact ordered-defect product never failed.
7. The shared algebra is verified symbolically: the four transition weights sum identically to one, the step objective equals $\text{const} + sD + h(s)$ so a common additive shift of the continuation values leaves D unchanged, and $\max_s \{sD + h(s)\} = \log_2(1 + 2^D)$ at $s = 1/(1 + 2^{-D})$ with second derivative $1/(s(s-1) \ln 2) < 0$.
8. The powers themselves are instantiated for $k \leq 3$ and re-audited from the constructed rows. At $k = 3$ the family has 91,125 members, all 21 coordinate counts equal the cap bound 36,450, incidence 765,450 exceeds the Reimer threshold 750,668, the columns are active and pairwise distinct, and a brute-force count over all 8,303,765,625 ordered pairs finds 8,227,000,000 missing joins, i.e. exactly $526528/531441 = 1 - (17/81)^3$. That count uses no product shortcut.

Complete smaller-dimension controls remain separate from the theorem. The full labeled $n = 4$ scan (which also contains lower-dimensional families via zero columns) retained 358 nontrivial cap-2/5 families. An exact $n = 5$ producer represented all 463,343 nontrivial admissible labeled families by 5,172 coordinate orbits; a resumable 120-order evaluation found minimum $A_+ = 0.006474313148642441 \dots$. At $n = 6$, exhaustive coverage is feasible only for declared

subclasses: all 262,585 admissible families with $3 \leq |\mathcal{F}| \leq 7$ (1,081 orbits), and all 450 orbits in the three two-block classes. Enumeration and defects in these controls are exact; their objective comparisons are float64. None is used to infer the all- k result.

A complete declared-class $n = 8$ search provides a finite control. It exhausts all 65,535 families invariant under $S_1 \times S_7$, finds 136 admissible full S_8 orbits (130 normalized), and finds two negative values. Its largest float64 ratio is $0.034335490331443\dots$, below the certified seven-coordinate base. This is a complete numerical result for that symmetry class, not an arbitrary- $n = 8$ optimum.

6 Scope and limitations

The sequence satisfies $\varepsilon_{\vee}(\mathcal{F}_k) \rightarrow 1$. This is no limitation for the unrestricted supremum (5), but it means the argument does not settle a different local-stability problem restricted to $\varepsilon_{\vee} \rightarrow 0$. Approximate-union-closure results such as Chase–Lovett [1] concern that small-defect regime and frequency bounds, not the present Bellman functional.

The base is deliberately non-union-closed. The theorem neither refutes the union-closed sets conjecture nor supplies the UC-specific structural charge needed by the larger entropy programme. Its conclusion is narrower and complete: no fixed scalar multiple of the ordered-pair closure defect can repair A_+ over all cap/Reimer families.

Both base certificates are verified interval computation, not proof-assistant artifacts, and the tensorization lemmas are human-audited mathematics supported by the falsification searches above. The conclusion depends on the numerics only through the sign of $A_+(\mathcal{B})$, which two arithmetically disjoint checkers establish: a 256-bit Arb computation with clamp classification, and a standard-library dyadic computation that avoids clamp decisions entirely and needs no third-party package.

References

- [1] Z. Chase and S. Lovett, *Approximate union closed conjecture*, arXiv:2211.11689 (2022).